Ofqual QAN	**600/6407/9**
Quartz ID	**1047**
Qualification	**Level 3 Diploma in Computing**

Assessment	**Mathematical Skills for Computing**
Assessment Type	**Examination**
Date	**5th March 2025**
Time	**09.00am – 12.00pm**

Marking Scheme

This marking scheme has been prepared as a **guide only** to markers. This is not a set of model answers, or the exclusive answers to the questions, and there will frequently be alternative responses which will provide a valid answer. Markers are advised that, unless a question specifies that an answer be provided in a particular form, then an answer that is correct (factually or in practical terms) **must** be given the available marks.

If there is doubt as to the correctness of an answer, the relevant NCC Education materials should be the first authority.

Throughout the marking, please credit any valid alternative point.

Answer ALL questions

Question 1 **Marks 1**

Simplify: $r^5 \times r^{-6} \times r^3$

a. r^{14}

b. r^{-90}

c. r^2

d. r^{-2}

Mark scheme
C

Question 2 **Marks 1**

Simplify the expression:

$$\frac{12\ a^2b^5}{4a^5\ b}$$

a. $\frac{3b^4}{a^3}$

b. $\frac{8\ b^4}{a^3}$

c. $\frac{3b^6}{a^7}$

d. $8\ a^2b^5a^{-5}b^{-1}$

Mark scheme
A

Question 3 **Marks 1**

Simplify the expression:

$$(3c^2c^3)^3$$

a. $9c^{18}$

b. $27c^{125}$

c. $27c^{15}$

d. $27c^8$

Mark scheme
C

Question 4 **Marks 1**

Simplify the expression:

$$e(3e + f) - 7e(f - e)$$

a. $4e^2 + 6ef$

c. $e^2 - 6ef$

© NCC Education Limited 2025

b. $10e^2 + 6ef$

d. $10e^2 - 6ef$

Mark scheme
D

Marks

Question 5 **1**

A taxi company charges a £3 initial hire fee and £2.50 for each mile travelled, m. Select the formula which can be used to calculate the cost, C, of a taxi ride.

a. $C = 3m + 2.50$

c. $C = 3 + 250m$

b. $C = 3 + 2.50m$

d. $C = 3m + 250$

Mark scheme
B

Marks

Question 6 **1**

If $C = \{2, 5, 10, 15, 23\}$ and $D = \{2, 10\}$, find $C - D$

a. $\{5, 15, 23\}$

c. $\{0, 5, 15, 23\}$

b. $\{0, 5, 0, 15, 23\}$

d. $\{5, 23\}$

Mark scheme
A

Marks

Question 7 **1**

If $A = \{1, 5, 9, 10\}$ and $B = \{1, 9\}$, which of the following is true

a. $A = B$

c. $A \subset B$

b. A is equivalent to B

d. $B \subset A$

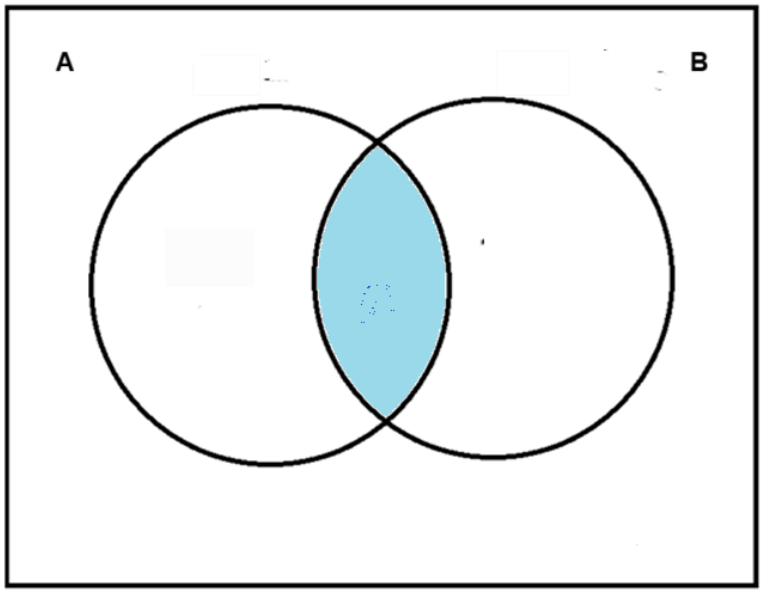
Mark scheme
D

Marks

© NCC Education Limited 2025

Question 8 **1**

Select the correct set notation for the shaded area of the following Venn diagram.



a. $A' \cap B'$

c. $A \cup B$

b. $A \cap B$

d. $A' \cup B'$

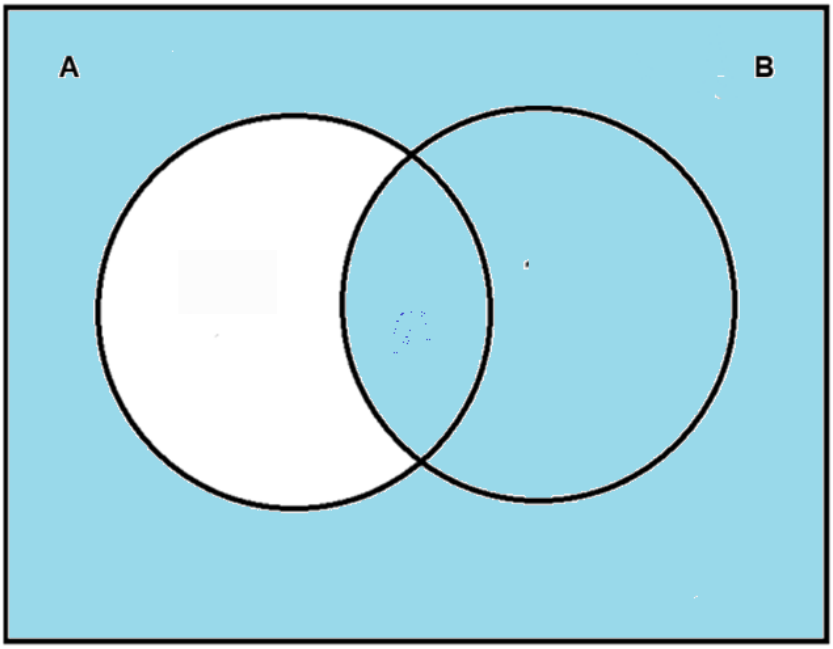
Mark scheme
B

© NCC Education Limited 2025

Marks

Question 9 **1**

Select the correct set notation for the shaded areas of the following Venn diagram.



a. $A' \cup B$

b. $A \cup B'$

c. $A' \cap B$

d. $A \cap B'$

Mark scheme

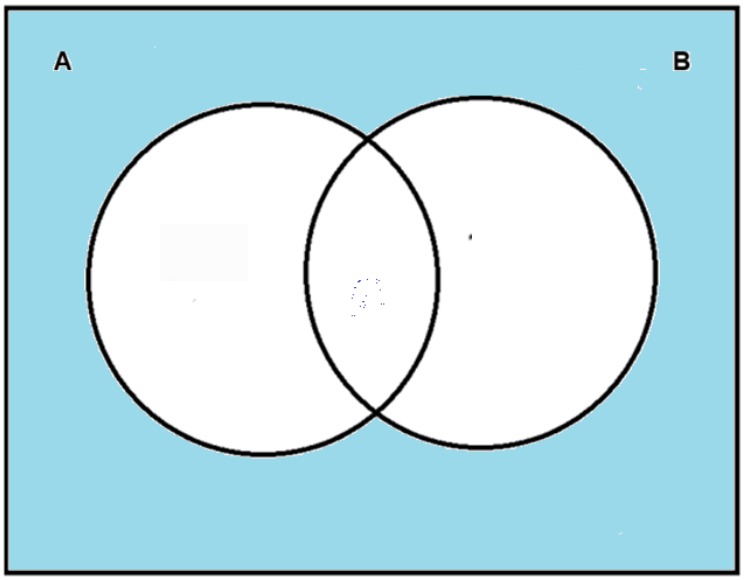
A

© NCC Education Limited 2025

Marks

Question 10 **1**

Select the correct set notation for the shaded area of the following Venn diagram.



a. $A' \cup B'$

c. $A' \cap B'$

b. $A \cup B$

d. $A \cap B$

Mark scheme

C

Marks

Question 11 **1**

Select the option that gives the order of the following matrix.

$$\begin{pmatrix} -2 & 3 \\ 0 & 4 \\ 3 & -1 \end{pmatrix}$$

a. 5

c. 6

b. 2 x 3

d. 3 x 2

Mark scheme

D

© NCC Education Limited 2025

Marks

Question 12 **1**

Name the type of matrix shown here:

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

a. Scalar matrix

b. Triangular matrix

c. Null matrix

d. Identity matrix

Mark scheme

C

Marks

Question 13 **1**

TWO (2) matrices are given as:

$$E = \begin{pmatrix} 3 & 1 \\ -2 & 4 \end{pmatrix} \text{ and } F = \begin{pmatrix} 2 & 0 \\ 1 & -1 \end{pmatrix}$$

Calculate $E - F$

a. $\begin{pmatrix} -1 & -1 \\ -3 & -5 \end{pmatrix}$

b. $\begin{pmatrix} 1 & 1 \\ -3 & 5 \end{pmatrix}$

c. $\begin{pmatrix} 5 & 1 \\ -1 & 3 \end{pmatrix}$

d. $\begin{pmatrix} -5 & -1 \\ 1 & -3 \end{pmatrix}$

Mark scheme

B

Marks

Question 14 **1**

If $E = \begin{pmatrix} 3 & 1 \\ -2 & 4 \end{pmatrix}$

Calculate EI

a. $\begin{pmatrix} 4 & -1 \\ -3 & 5 \end{pmatrix}$

b. $\begin{pmatrix} -3 & -1 \\ 2 & -4 \end{pmatrix}$

c. $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

d. $\begin{pmatrix} 3 & 1 \\ -2 & 4 \end{pmatrix}$

Mark scheme

D

© NCC Education Limited 2025

Question 15 **Marks 1**

Find the determinant of F if $F = \begin{pmatrix} 2 & 0 \\ 1 & -1 \end{pmatrix}$

a. 1

b. -1

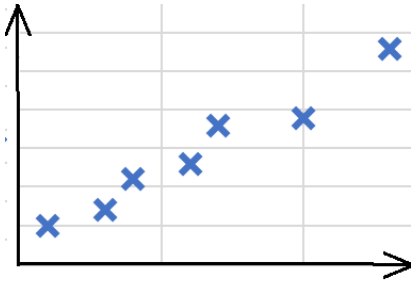
c. 2

d. -2

Mark Scheme
D

Question 16 **Marks 1**

Select the option which best describes the type of correlation shown in the scatter graph below.



a. Strong positive correlation.

b. Weak negative correlation

c. No correlation

d. Strong negative correlation

Mark scheme
A

Question 17 **Marks 1**

State which of the following data, collected during a consumer survey, is continuous data.

a. Number of times a person visits a supermarket each month.

b. Preferred supermarket

c. Amount of time spent in the supermarket

d. Method of transport used to travel to the supermarket

Mark scheme
C

© NCC Education Limited 2025

Marks

Question 18 **1**

The amount of time taken by a group of people to solve a puzzle is recorded in the stem and leaf diagram below.

0	9
1	1 1 3 5 9
2	0 0 0 2 6 7
3	2 8 9
4	0

- N = 16 Key: 3 | 2 means 32

Calculate the range of the time taken to solve the puzzle.

a. 9

c. 16

b. 31

d. 40

Mark scheme
B

Marks

Question 19 **1**

Using the same stem and leaf diagram as in Question 18 and reproduced below:

0	9
1	1 1 3 5 9
2	0 0 0 2 6 7
3	2 8 9
4	0

N = 16 Key: 3 | 2 means 32

Calculate the mode of the time taken to solve the puzzle.

a. 20

c. 40

b. 11

d. 9

Mark scheme
A

© NCC Education Limited 2025

Marks

Question 20 **1**

Using the same stem and leaf diagram as in Question 18 and reproduced below:

0	9
1	1 1 3 5 9
2	0 0 0 2 6 7
3	2 8 9
4	0

N = 16 Key: 3 | 2 means 32

Find the number of people who took 9 minutes to solve the puzzle.

a. 0

c. 1

b. 2

d. 9

Mark scheme
C

Marks

Question 21 **1**

A data set is given by M = {92, 73, 42, 32, 32, 41, 80, 24}
Calculate the mean of the data set.

a. 92

c. 32

b. 52

d. 24

Mark scheme
B

Marks

Question 22 **1**

A survey asks a group of people the amount of time it takes them to travel to work.
Select the option below which best describes the data collected:

a. Qualitative

c. Discrete

b. Secondary

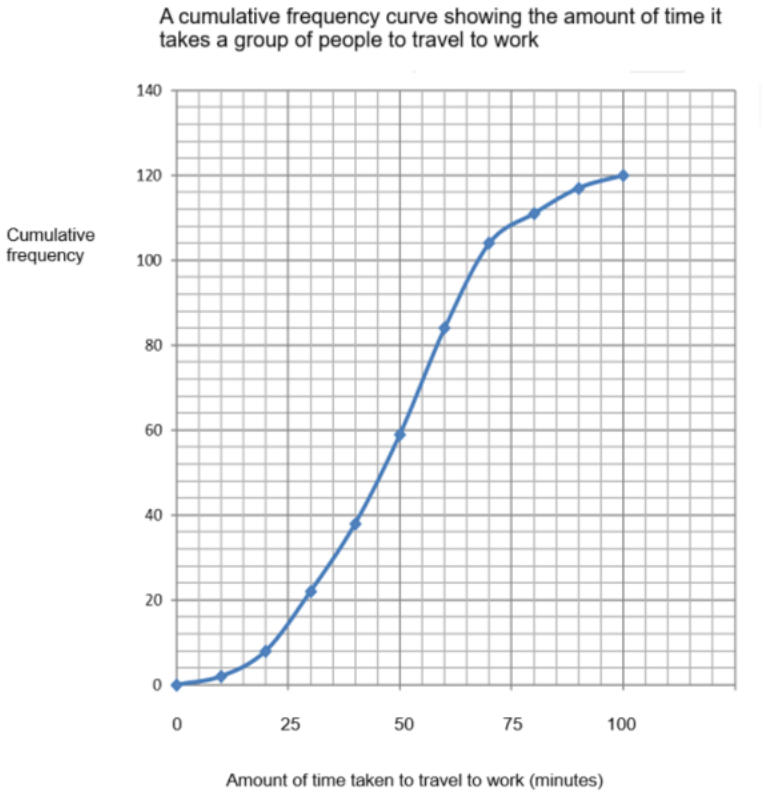
d. Primary

Mark scheme
D

© NCC Education Limited 2025

Question 23 **Marks 1**

The data from the survey in Question 22 is displayed in the cumulative frequency curve below.



Find the number of people who were surveyed.

a. 140

c. 120

b. 100

d. 20

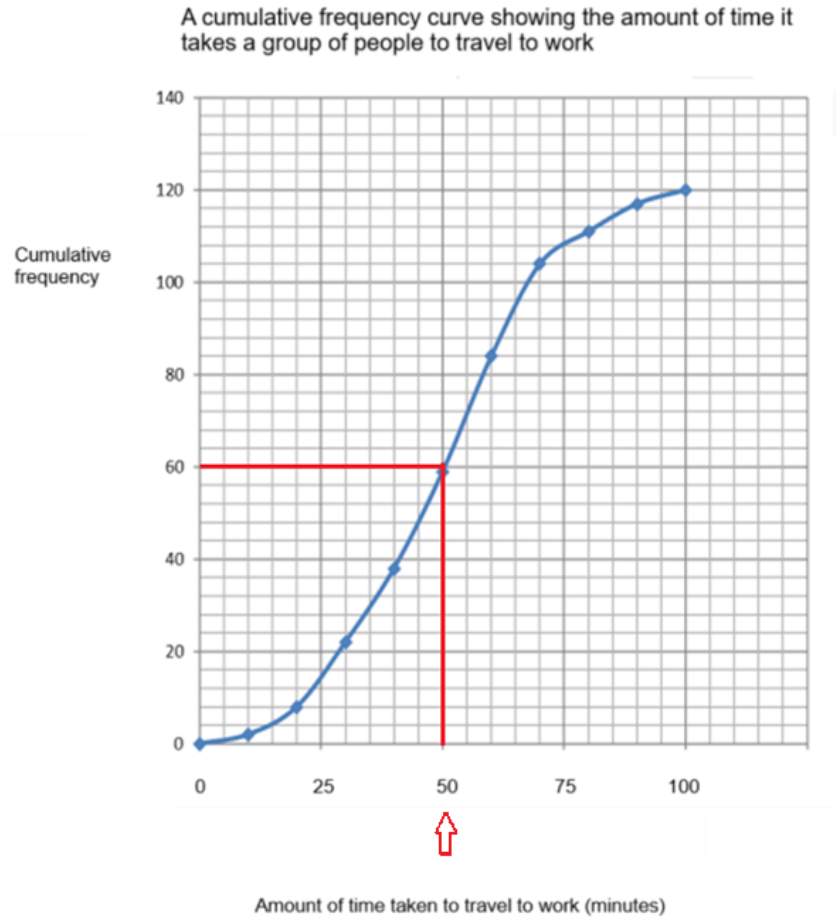
Mark scheme
C

© NCC Education Limited 2025

Marks

Question 24 **1**

Select the statistical measure that is indicated by the red arrow on the cumulative frequency below:



a. Median

b. Mean

c. Mode

d. Range

Mark scheme

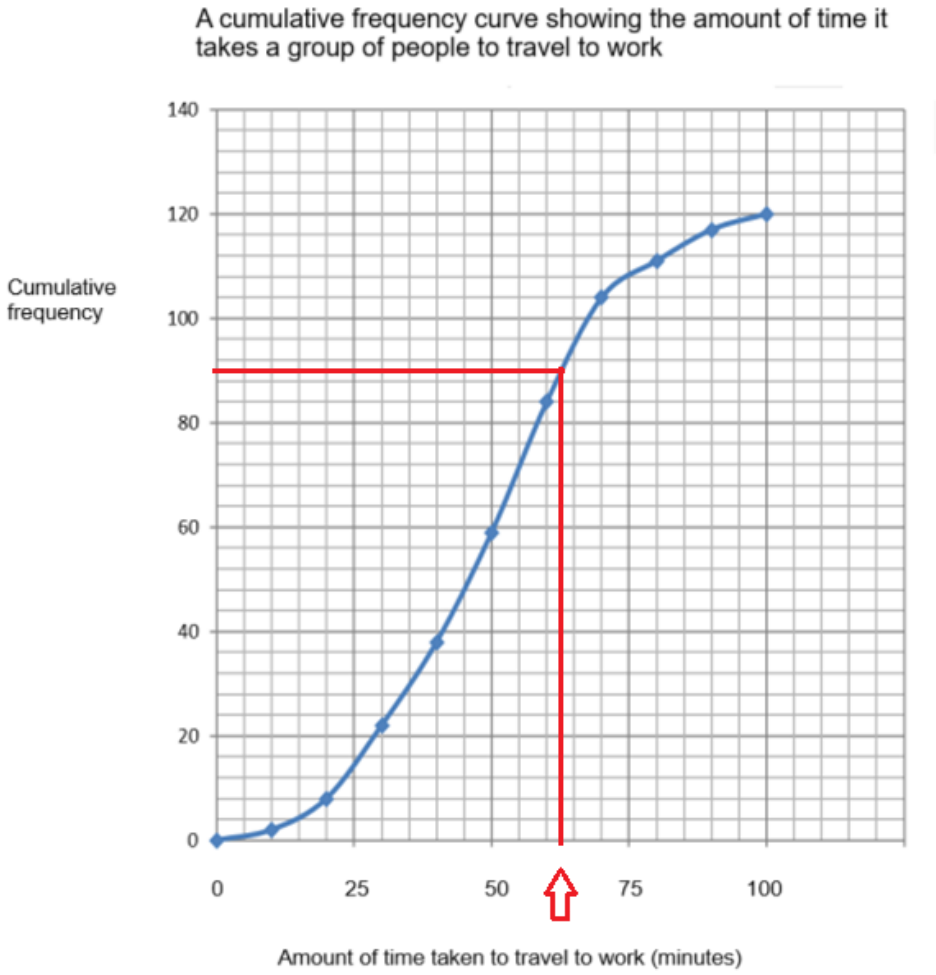
A

© NCC Education Limited 2025

Marks

Question 25 **1**

Select the statistical measure that is indicated by the red arrow on the cumulative frequency below:



a. Median
b. Lower quartile
c. Upper quartile
d. Interquartile range

Mark scheme

C

© NCC Education Limited 2025

Marks

Question 26 **1**

The letters of the word STATISTICS are placed as individual tiles in a bag.
One tile is drawn out at random.
Calculate the probability of the tile being the letter ‘I’.

a. 0.2

c. 1

b. 0.1

d. 2

Mark scheme
A

Marks

Question 27 **1**

The letters of the word STATISTICS are placed as individual tiles in a bag.
One tile is drawn out at random.
Calculate the probability of the tile **NOT** being the letter ‘I’.

a. 0.9

c. 0.8

b. 0.1

d. 8

Mark scheme
C

Marks

Question 28 **1**

The probability that a lettuce seed will germinate is 0.95.
120 lettuce seeds are planted.
Estimate the number of lettuce seeds will germinate.

a. 95

c. 120

b. 114

d. 25

Mark scheme
B

© NCC Education Limited 2025

Marks

Question 29 **1**

A college records the number of times a group of students are late each month:

Number of times students are late.	0	1	2	3	4	5 or more
Number of students	62	21	10	4	2	1

Calculate the relative frequency of a student being late twice this month.

a. 10

b. $\frac{1}{10}$

c. 2

d. 0.2

Mark scheme
B

Marks

Question 30 **1**

A college records the number of times a group of students are late each month:

Number of times students are late.	0	1	2	3	4	5 or more
Number of students	62	21	10	4	2	1

Calculate the relative frequency of a student being late three or four times this month.

a. $\frac{6}{10}$

b. $\frac{3}{50}$

c. 6

d. 0.6

Mark scheme
B

Total 30 marks

© NCC Education Limited 2025

Question 31

a) Completely factorise: **2**

$$4a^5b^2 - 8b^7$$

Mark scheme

$\boldsymbol{4b^2(a^5 - 2b^5)}$

1 mark for $4b^2$, 1 mark for $(a^5 - 2b^5)$

b) Simplify: **2**

$$\frac{5}{c} + \frac{3}{4c}$$

Mark Scheme

$= \frac{\mathbf{20+3}}{\mathbf{4c}}$

$= \frac{\mathbf{23}}{\mathbf{4c}}$

1 mark for correct method, 1 mark for correct answer.

c) Solve the following simultaneous equations algebraically and find the value of x and y: **3**

$$5x + y = -7 \text{ and } y = 4x + 11$$

Mark scheme

Substitute the second equation into the first equation:

$$\mathbf{5x + (4x + 11) = -7}$$
$$\mathbf{9x + 11 = -7}$$
$$\mathbf{9x = -18}$$

Hence,

$$\boldsymbol{x = -2}$$

Substitute $x = -2$ into the second equation

$$\mathbf{y = 4(-2) + 11}$$
$$\mathbf{y = 3}$$

1 mark for correct method, 1 mark for correct value of y and 1 mark for correct value of x.

d) Solve the quadratic equation by factorising: **3**

$$x^2 + 3x - 40 = 0$$

© NCC Education Limited 2025

Mark scheme

Rewrite as

$$x^2 + 3x - 40 = 0$$
$$(x + 8)(x - 5) = 0$$

So,

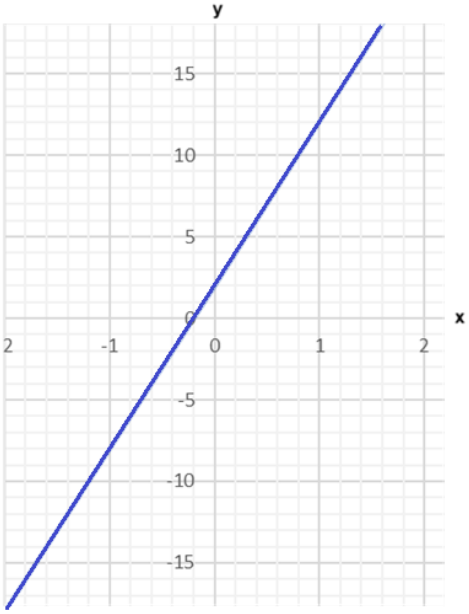
$$x = -8 \text{ or } x = 5$$

1 mark for $(x + 8)$
1 mark for $(x - 5)$
1 mark for correct values of x.

Total 10 Marks

Question 32

a) Determine the equation of the line shown below. **2**



Mark scheme
$y = 10x + 2$
Award 1 mark for $10x$ ***and 1 mark for*** $+2$

b) Copy and complete the table of values for the curve $y = 2x + 6 - x^2$ for $-2 \leq x \leq 4$ **3**

x	-2	-1	0	1	2	3	4
$y = 2x + 6 - x^2$							

© NCC Education Limited 2025

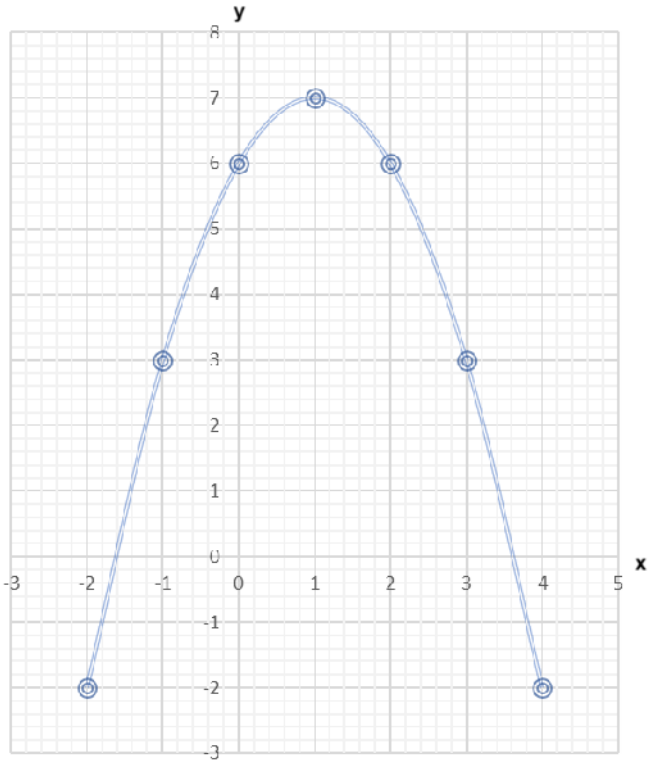
Mark scheme

Table of values

x	-2	-1	0	1	2	3	4
$y = 2x + 6 - x^2$	-2	3	6	7	6	3	-2

Award 1 mark for TWO (2) to THREE (3) correct values,
2 marks for FOUR (4) to SIX (6) correct values,
3 marks for all values correct

c) Use the values from (b) to plot a graph of the curve $y = 2x + 6 - x^2$ **4**
You should use the graph paper provided.
Mark Scheme



1 mark for correctly labelled axes and scales.

2 marks for correctly plotted points.
Deduct 1 mark per error up to a maximum of 2 marks.

1 mark for points connected by a smooth curve.

© NCC Education Limited 2025

d) Use the graph constructed in part (c) to solve the equation $x^2 = 2x + 6$ **1**

Mark scheme

$\boldsymbol{x = -1.6}$ ***or*** $\boldsymbol{x = 3.6}$

Accept values of $\boldsymbol{x = -1.6 \pm 0.1}$ ***and*** $\boldsymbol{x = 3.6 \pm 0.1}$

Award 1 mark only if both values of x are given.

Total 10 Marks

Question 33

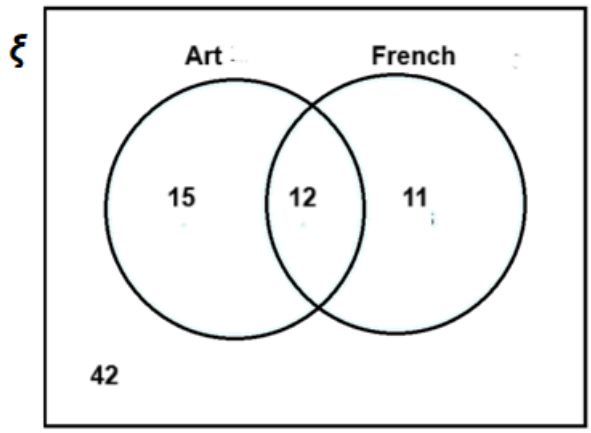
a) A community centre runs two classes on a Wednesday, an arts class and a French class.

80 people visit the community centre on Wednesday.
38 people attend at least one of the classes.
27 people attend the art class.
12 people attend both the art class and the French class.

i) Represent this information on a Venn diagram. **4**

© NCC Education Limited 2025

Mark scheme



Award 3 marks for values in all four regions correct.
Award 2 marks for values in two or three of the four regions correct.
Award 1 mark for value in one of the four regions correct.

Award 1 mark for all labels correct.

Total 4 marks.

ii) A person who attended the community centre on Wednesday is selected at random. Find the probability that they attended the French class only. **1**
Mark scheme

Number of people who attended the French class only is 11. So, the probability is $\frac{11}{80}$ ***(1 mark) If answer incorrect but consistent with the diagram award 1 mark as FT)***

iii) A person who attended the community centre on Wednesday is selected at random. Find the probability that they attended exactly one class. Simplify your answer. **2**
Mark Scheme

Probability that they attended exactly one class is:

$$\frac{11+15}{80} = \frac{26}{80} = \frac{13}{40}$$

(1 mark for workings, 1 mark for correct answer, simplified as far as possible. If answer incorrect but consistent with the diagram, award 1 mark as FT)

© NCC Education Limited 2025

b) If the universal set $U = \{3,\ 6,\ 9,\ 12,\ 15,\ 18\}$ and set $A = \{12, 15, 18\}$:

i) Find the complement of set A. **1**

Mark scheme

$\boldsymbol{A' = \{3, 6, 9\}}$

(1 mark)

ii) State whether A is an infinite or finite set. **1**

Mark scheme

Finite

(1 mark)

iii) Find $n(A)$. **1**

Mark scheme

$\boldsymbol{n(A) = 3}$

(1 mark)

Total 10 Marks

Question 34

a) TWO (2) matrices are given as:

$$A = \begin{pmatrix} -4 & 5 \\ 2 & 7 \end{pmatrix} \text{ and } B = \begin{pmatrix} 3 & 4 \\ -2 & 2 \end{pmatrix}$$

i) Calculate $A + B$. **1**

Mark scheme

$$\boldsymbol{A + B = \begin{pmatrix} -1 & 9 \\ 0 & 9 \end{pmatrix}}$$

(1 mark for all elements correct)

ii) Calculate $-3A$. **1**

Mark scheme

$$\boldsymbol{-3A = \begin{pmatrix} 12 & -15 \\ -6 & -21 \end{pmatrix}}$$

(1 mark for all elements correct)

iii) State the transpose of matrix B **1**

Mark scheme

$$\boldsymbol{B^T = \begin{pmatrix} 3 & -2 \\ 4 & 2 \end{pmatrix}}$$

(1 mark for all elements correct).

iv) Find A^2 **2**

© NCC Education Limited 2025

Mark scheme

$$\boldsymbol{A^2 = AA} = \begin{pmatrix} -4 & 5 \\ 2 & 7 \end{pmatrix}\begin{pmatrix} -4 & 5 \\ 2 & 7 \end{pmatrix} = \begin{pmatrix} (-4 \times -4) + (5 \times 2) & (-4 \times 5) + (5 \times 7) \\ (2 \times -4) + (7 \times 2) & (2 \times 5) + (7 \times 7) \end{pmatrix}$$

$$= \begin{pmatrix} 26 & 15 \\ 6 & 59 \end{pmatrix}$$

(1 mark for TWO (2) or THREE (3) elements correct, 2 marks for all elements correct.)

b) Given that **3**

$$C = \begin{pmatrix} -4 & 3 \\ 0 & 2 \end{pmatrix}$$

show that C has an inverse and find C^{-1}

Mark scheme

The determinant of C is given by:

$\boldsymbol{(-4 \times 2) - (3 \times 0) = -8}$ ***·(1 mark)***

Therefore, determinant of C ≠ 0 and so has an inverse (1 mark)

$$\boldsymbol{C^{-1} = -\frac{1}{8}\begin{pmatrix} 2 & -3 \\ 0 & -4 \end{pmatrix}}$$

$$= \begin{pmatrix} -\frac{1}{4} & \frac{3}{8} \\ 0 & \frac{1}{2} \end{pmatrix}$$

(1 mark for all elements correct).

c) Find the matrix that represents the linear transformation: **2**

$$D: \begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} -3y \\ -2x + 4y \end{pmatrix}$$

© NCC Education Limited 2025

Mark scheme

$$\begin{pmatrix} -3y \\ -2x + 4y \end{pmatrix} = \begin{pmatrix} 0x - 3y \\ -2x + 4y \end{pmatrix}$$

Therefore, the transformation matrix is

$$\begin{pmatrix} 0 & -3 \\ -2 & 4 \end{pmatrix}$$

1 mark for TWO (2) or THREE (3) elements correct, 2 marks for all elements correct.

Total 10 Marks

© NCC Education Limited 2025

Question 35

The marks achieved by a group of 23 students at Greenville College in an IT exam is as follows

38 41 53 53 54 54 57 62 68 68 68 68 71 71 72 72 74 74
74 87 88 89 92

a) State whether the data collected is quantitative or qualitative. **1**
Mark scheme
Quantitative (1 mark)

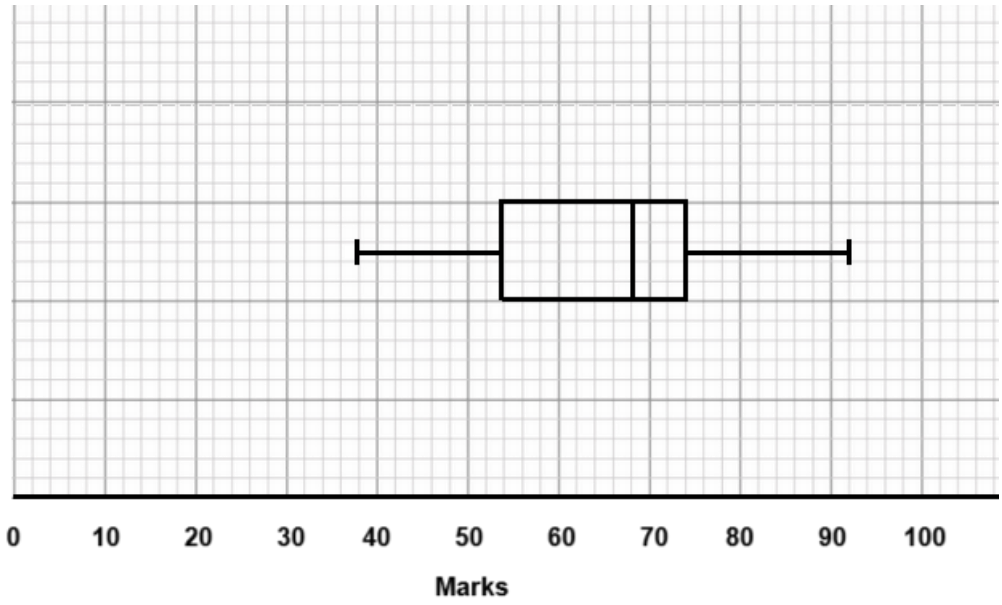
b) Find the median value of the data. **1**
Mark scheme
Median value is 68(1 mark)

c) Find the upper quartile and lower quartile of the data. **2**
Mark scheme
Upper quartile is 74 (1 mark)
Lower quartile is 54 (1 mark)

d) Construct a box plot for the data. Use the graph paper provided. **4**

© NCC Education Limited 2025

Mark scheme



Award 1 mark for correct upper and lower bounds
Award 1 mark for correct upper and lower quartile, i.e. interquartile range
Award 1 mark for correct median.

Award 1 mark for correctly labelled horizontal axis.

e) A group of students from Central College sits the same exam. **2**
The median mark achieved by the Central College students is 73
The interquartile range is 25 and the range is 61.
Compare the data for the Central College results with those of the Greenville College results.

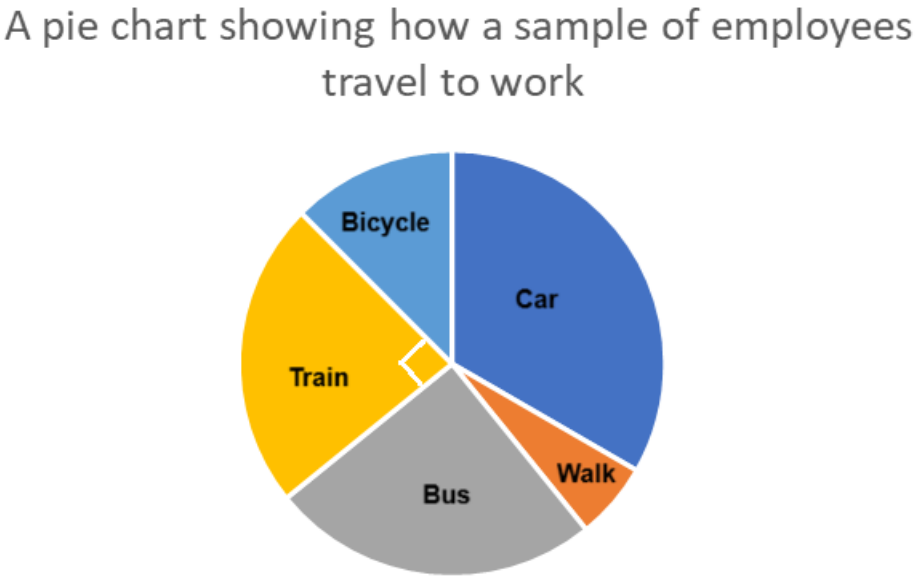
Mark scheme
The median average mark is higher for the Central College results than the Greenville College results (1 mark).
However, the marks for Greenville College are more consistent than the Central College results (1 mark).

Total 10 Marks

© NCC Education Limited 2025

Question 36

a)



i) State the least popular method of transport to work. **1**

Mark scheme

Walk

(1 mark)

ii) State the fraction of the employees who use the train to get to work. **1**

Mark scheme

¼

(1 mark)

iii) 30 employees travel to work by train. Calculate the total number of employees in the sample. **2**

Mark scheme

$$\mathbf{30 \times 4 = 120}$$

1 mark for correct method, 1 mark for correct value

© NCC Education Limited 2025

b) The amount of time it takes a sample of employees to travel to work each day is recorded in the table below.

Time, t (minutes)	Frequency
$0 \leq t < 10$	1
$10 \leq a < 20$	7
$20 \leq a < 30$	17
$30 \leq a < 40$	30
$40 \leq a < 50$	42
$50 \leq a < 60$	22
$60 \leq a < 70$	0
$70 \leq a < 80$	1

i) Calculate an estimate of the mean of the data. State the level of accuracy to which you have rounded your answer. **4**

Mark scheme

Time, t (minutes)	***Midpoint***	Frequency	$\boldsymbol{fx}$
$0 \leq t < 10$	***5***	1	***5***
$10 \leq a < 20$	***15***	7	***105***
$20 \leq a < 30$	***25***	17	***425***
$30 \leq a < 40$	***35***	30	***1050***
$40 \leq a < 50$	***45***	42	***1890***
$50 \leq a < 60$	***55***	22	***1210***
$60 \leq a < 70$	***65***	0	***0***
$70 \leq a < 80$	***75***	1	***75***

1 mark for fx values.
1 mark for $\Sigma fx = 4760$

$$\bar{x} = \frac{\Sigma fx}{\Sigma f}$$

$$= \frac{4760}{120}$$ ***(1 mark)***

$$= 39.7$$ ***minutes (to 3 s.f.) (1 mark).***

ii) Explain whether you would use the median or the mean to calculate the average amount of time it takes employees to travel to work. You should refer to a specific value within the data set. There is no single correct answer for this question but you must justify your choice. **2**

© NCC Education Limited 2025

Mark scheme

Candidates may select either the median or the mean but must justify their choice. For example:
Mean (1 mark) because it takes into consideration all the values including any outliers such as the one employee who takes between 70 and 80 minutes to travel to work (1 mark)
Alternatively,
Median (1 mark). This gives a more typical value as it excludes any outliers such as the one employee who takes between 70 and 80 minutes to travel to work (1 mark).

Total 10 Marks

© NCC Education Limited 2025

Question 37

a) The two-way table shows the favourite sport of 180 students.

	Cricket	Soccer	Other	Total
Female	48	23		102
Male			21	
Total		55	52	180

i) Complete the two way table. **3**

Mark scheme

	Cricket	Soccer	Other	Total
Female	48	23	**31**	102
Male	***25***	***32***	21	**78**
Total	***73***	55	52	180

(1 mark for two correct values, 2 marks for three or four correct values, 3 marks for five correct values.)

ii) A student is selected at random. Write down the probability that their favourite sport is soccer. Give your answer as a fraction in its simplest form. **1**

Mark scheme

$$\frac{\mathbf{55}}{\mathbf{180}} = \frac{\mathbf{11}}{\mathbf{36}}$$

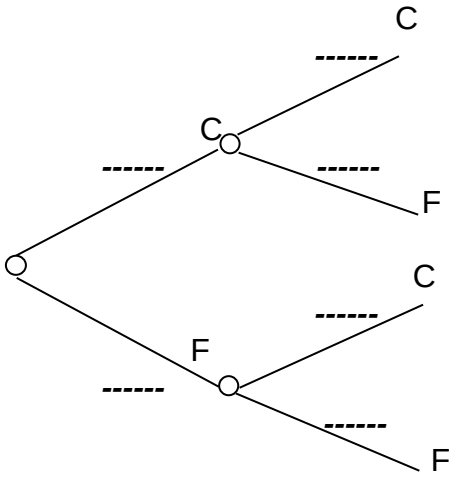
(1 mark for correct answer simplified, if answer incorrect but consistent with answer to (a)(i) award 1 mark as FT)

b) A bakery sells cakes. SEVEN (7) of the cakes are chocolate and THREE (3) of the cakes are fruit.
TWO (2) cakes are selected at random (without replacement).

© NCC Education Limited 2025

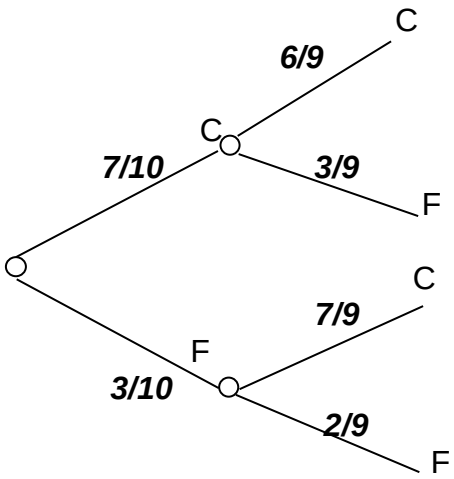
i) Copy and complete the probability tree diagram where C represents chocolate cake and F represent fruit cake. **3**

Selection 1 Selection 2



Mark scheme

Selection 1 Selection 2



1 mark for correct workings 7/10 and 3/10 on first branch. Award 1 mark for 6/9 and 3/9 on upper branch for selection 2 and 1 mark for 7/9 and 2/9 on lower branch for selection 2; maximum of 3 marks.

ii) Calculate the probability that **at least one** of the cakes selected is chocolate. Write your answer as a fraction. **3**

© NCC Education Limited 2025

Mark scheme
P(at least one chocolate) = 1 – P(FF)

$= \mathbf{1} - (\frac{\mathbf{3}}{\mathbf{10}} \times \frac{\mathbf{2}}{\mathbf{9}})$ ***(1 mark method)***

$= \mathbf{1} - \frac{\mathbf{1}}{\mathbf{15}}$ ***(1 mark)***

$= \frac{\mathbf{14}}{\mathbf{15}}$ ***(1 mark correct answer simplified or not)***

If method correct but value incorrect due to errors in (i) award 1 mark for method and 0 marks for answer. Marks should not be awarded for a rounded answer.
Candidates may use longer, alternate method, i.e, P(CF)+P(FC)+P(CC) and they should be awarded the mark for method.

Total 10 Marks

End of Paper

© NCC Education Limited 2025

Learning Outcomes matrix

Question	Learning Outcomes assessed	Marker can differentiate between varying levels of achievement
1	1	Yes
2	1	Yes
3	1	Yes
4	1	Yes
5	2	Yes
6	4	Yes
7	4	Yes
8	4	Yes
9	4	Yes
10	4	Yes
11	5	Yes
12	5	Yes
13	5	Yes
14	5	Yes
15	5	Yes
16	6	Yes
17	6	Yes
18	6	Yes
19	6	Yes
20	6	Yes
21	6	Yes
22	6	Yes
23	6	Yes
24	7	Yes
25	7	Yes
26	8	Yes
27	8	Yes
28	8	Yes
29	8	Yes
30	8	Yes
31	1, 2	Yes
32	3	Yes
33	4	Yes
34	5	Yes
35	6, 7	Yes
36	7	Yes
37	8	Yes

© NCC Education Limited 2025

Grade descriptors

Learning Outcome	Pass	Merit	Distinction
Be able to perform a range of algebraic calculations	Demonstrate ability to perform calculations	Demonstrate ability to perform calculations consistently well	Demonstrate ability to perform all calculations to the highest standard
Be able to solve a range of basic equations	Demonstrate ability to perform techniques	Demonstrate ability to perform techniques consistently well	Demonstrate ability to perform techniques to the highest standard
Be able to represent and solve algebraic equations through graphical solutions	Demonstrate ability to perform techniques	Demonstrate ability to perform techniques consistently well	Demonstrate ability to perform techniques to the highest standard
Understand the fundamentals of Set Theory	Demonstrate adequate understanding of techniques	Demonstrate robust understanding of techniques	Demonstrate highly comprehensive understanding of techniques
Understand the fundamentals of Matrices	Demonstrate adequate understanding of techniques	Demonstrate robust understanding of techniques	Demonstrate highly comprehensive understanding of techniques
Be able to present data and relationships in graphical form	Demonstrate adequate understanding of techniques	Demonstrate robust understanding of techniques	Demonstrate highly comprehensive understanding of techniques
Understanding and use simple descriptive statistics	Demonstrate adequate understanding of techniques	Demonstrate robust understanding of techniques	Demonstrate highly comprehensive understanding of techniques
Understand the fundamentals of Probability	Demonstrate adequate understanding of techniques	Demonstrate robust understanding of techniques	Demonstrate highly comprehensive understanding of techniques

© NCC Education Limited 2025